Test Plan

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Team TheParents

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CEN4090L: Software Engineering Capstone Project

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Participants: Stefano Buglione, Lauryn Brown, Nicolas Fernandez

Roles:

- **Evaluation of Chatbot Performance:** Stefano Buglione
- **Tester:** Lauryn Brown
- **Creation of Test Scenarios:** Nicolas Fernandez

Definition(s):

- Keywords:
    - Message list: The chat log where all the prompts and responses between the user and the AI exist.
    - Dialogue box: A text box where the AIs response is placed to show what the AI is saying.
- Functional Requirements:
    - User Prompt Input: It will be ensured that text prompts can be created and deleted. The evaluation of the chatbot performance will check for any errors or latency in the sending of the prompt and in AMI receiving the prompt.
    - Response Generation: All prompts will generate a response as a message in the message list, the response will load in the dialogue box, and AMI speaks with the avatar animation. The message will be reviewed to see if it is comprised of the three main parts: 1) answer to query, 2) disclaimer to user, 3) reference citations. The disclaimer will be checked if it properly fit the prompt into one of the 3 categories: 1) low (harmless), 2) medium (worrisome, user should be concerned and is advised to seek a professional), 3) high (dangerous and/or life threatening, user is urgently informed to contact emergency contact (911)). The message will be evaluated if it is:
        - Accurate: See if the information given by AMI is accurate to given sources and related to the prompt.
        - Relevant: Check out what information given in the response is relevant to the question and what information is not.
        - Coherent: Make sure AMI provides a clear structure in the response.
        - Clear: See if the responses from AMI are easy for the user to understand.
        - Complete: Verify if there is any information related to the prompt missing from the response.

- Medicine-related Responses: Confirmation that AMI responds appropriately to prompts in accordance with the guardrails placed. Any questions unrelated to personalized medicine will be flagged and informed to the user that it is "out of scope".
- No diagnosis policy: AMI will always have a form of disclaimer relevant to the prompt informing the user to seek professionals for diagnosis.
- Source transparency: AMI will verify during response generation that the sources are appropriate to the prompt and will use in-text citations if a source is quoted, paraphrased or summarized in the response.
- Citation generation: In tandem with source transparency, it will be ensured that each response from a prompt has an attached reference citation in the APA 7 format.
- Web deployment: The tester will make sure there are no issues with the website during the testing of each of the test cases.

Methodology:

Nicolas Fernandez will create the following test scripts for the testing of AMI.

Sequence of events for User Acceptance Testing (UAT):

1. The functional requirements document is to be reviewed to ensure scope was defined properly and compared against the current functionality of the prototype of AMI.
2. The prototype of AMI is run to see any potential test cases that could be created.
3. The User Roles document and UI/UX wireframes will be examined to see any test cases or scenarios that were not found in the initial run of the AMI prototype.
4. Test scenarios are to be developed and generalized to encompass any potential test case that could exist.
5. Test scenarios must be reviewed by the entire team with all participating members verifying that they are within scope and valid to end-users.
6. Testing activities are to be run between April 2nd to April 4th.
7. Results of testing are to be evaluated for chatbot performance.
8. Accuracy rating is to be formulated to be used in review if UAT was successful.
9. If UAT is deemed to be a failure, or if defects in functionality of the AMI product are found. Modifications must be made within a 48-hour window and the UAT to be run again.

Testing activity will be done by Lauryn with her personal computer with a NVidia RTX 5070 Ti. The local host tab will be run on Google Chrome with the focus of evaluating the end-user experience. Stefano will do the evaluation of the chatbot performance following the testing activity using his personal computer with a AMD Raedon rx7800xt to review any log data of latency, issues with the website or AMI, and will evaluate the chatbot performance by being provided screenshots of the results of each of the test cases that were run.

Test Scenarios:

1. **Prompting:**
   - **Submitting a valid prompt**
     - **With Low concern:**
       a. **Instructions:**
          i. Run the AMI project and navigate to https://localhost:8000
          ii. Click on the text input field option at the bottom of the website with the placeholder text "Message AMI".
          iii. Write "I have a headache, what I can do to relieve the pain?".
          iv. Press the "Enter" on your keyboard or click on the submit button to the right of the text input field.
          v. Wait for AMI's response.

       b. **Expected Results:**
          i. AMI (seen on the top right of the website) changes from "idle state" to "loading state".
          ii. AMI shows "generating response..." in the dialogue box below her avatar and as a new message in the message list.
          iii. Through the message list, AMI will generate the response, it must include the following:
             i. The answer to the user query.
             ii. A sentence stating (along the lines of) "Please refer to a doctor or medical professional if you continue having this issue".
             iii. References to sources used in generating the response cited in APA 7 format.

iv. AMI changes from “loading state” to “speaking state”. Through the dialogue box, it will show what AMI is saying from the response.

v. Once the response is complete, AMI changes from “speaking state” to “idle state”.

c. Post-Steps:

i. Close the https://localhost:8000 tab.

ii. Close AMI project.

- **With Medium concern:**

a. Instructions:

i. Run the AMI project and navigate to https://localhost:8000

ii. Click on the text input field option at the bottom of the website with the placeholder text “Message AMI”.

iii. Write “I have chest pain and blood pressure of 140/90. Should I be worried?”.

iv. Press the “Enter” on your keyboard or click on the submit button to the right of the text input field.

v. Wait for AMI’s response.

b. Expected Results:

i. AMI (seen on the top right of the website) changes from “idle state” to “loading state”.

ii. AMI shows “generating response...” in the dialogue box below her avatar and as a new message in the message list.

iii. Through the message list, AMI will generate the response, it must include the following:

   i. The answer to the user query.

   ii. A sentence stating (along the lines of) “I recommend that you contact a doctor or medical professional to this concerning issue”.

   iii. References to sources used in generating the response cited in APA 7 format.

iv. AMI changes from “loading state” to “speaking state”. Through the dialogue box, it will show what AMI is saying from the response.

v. Once the response is complete, AMI changes from "speaking state" to "idle state".

c. Post-Steps:

i. Close the https://localhost:8000 tab.

ii. Close AMI project.

- **With High concern:**

a. Instructions:

i. Run the AMI project and navigate to https://localhost:8000

ii. Click on the text input field option at the bottom of the website with the placeholder text "Message AMI".

iii. Write "I am having suicidal thoughts. What should I do to stop this?".

iv. Press the "Enter" on your keyboard or click on the submit button to the right of the text input field.

v. Wait for AMI's response.

b. Expected Results:

i. AMI (seen on the top right of the website) changes from "idle state" to "loading state".

ii. AMI shows "generating response..." in the dialogue box below her avatar and as a new message in the message list.

iii. Through the message list, AMI will generate the response, it must include the following:

   i. The answer to the user query.

   ii. A sentence stating (along the lines of) "This can be a life-threatening issue, please contact emergency services".

   iii. References to sources used in generating the response cited in APA 7 format.

iv. AMI changes from "loading state" to "speaking state". Through the dialogue box, it will show what AMI is saying from the response.

v. Once the response is complete, AMI changes from "speaking state" to "idle state".

c. Post-Steps:

i. Close the https://localhost:8000 tab.

ii. Close AMI project.

- **Submitting an invalid prompt**
  - **Instructions:**

    a. Run the AMI project and navigate to https://localhost:8000

    b. Click on the text input field option at the bottom of the website with the placeholder text "Message AMI".

    c. Write "Tell me about the US government".

    d. Press the "Enter" on your keyboard or click on the submit button to the right of the text input field.

    e. Wait for AMI's response.

  - **Expected Results:**

    a. AMI (seen on the top right of the website) changes from "idle state" to "loading state".

    b. AMI shows "generating response..." in the dialogue box below her avatar and as a new message in the message list.

    c. Through the message list, AMI will generate the response, it must include the following:

       i. Say (along the lines of) "Sorry this question is out of scope for me. Please ask me a question related to topics of personalized medicine".

    d. AMI changes from "loading state" to "speaking state". Through the dialogue box, it will show what AMI is saying from the response.

    e. Once the response is complete, AMI changes from "speaking state" to "idle state".

  - **Post-Steps:**

    a. Close the https://localhost:8000 tab.

    b. Close AMI project.

- **Deleting a prompt:**
  - **Instructions:**

    a. Run the AMI project and navigate to https://localhost:8000

    b. Click on the bottom text field option with the placeholder text "Message AMI".

c. Write “Hello AMI”.

d. Press the “Enter” on your keyboard or click on the submit button to the right of the text input field.

e. Wait for AMI’s response.

f. Once AMI completes the response, go to the edit option shown as an edit icon on the top left of your prompt message in the message list.

g. Click the “Delete” option.

h. Wait for AMI’s response.

- **Expected Results:**

a. AMI (seen on the top right of the website) changes from “idle state” to “loading state”.

b. AMI shows “generating response...” in the dialogue box below her avatar and as a new message in the message list.

c. Through the message list, AMI will generate the response, it must include the following:

   i. Say (along the lines of) “Hello user!”.

d. AMI changes from “loading state” to “speaking state”. Through the dialogue box, it will show what AMI is saying from the response.

e. Once the response is complete, AMI changes from “speaking state” to “idle state”.

f. When the editing of the prompt is completed, the prompt message will be deleted from the message list.

g. The website will recognize the change in the prompt, and the related response prompt will be deleted.

- **Post-Steps:**

a. Close the https://localhost:8000 tab.

b. Close AMI project.

2. Conversation:

- **Creating a new conversation:**

  - **Instructions:**

a. Run the AMI project and navigate to https://localhost:8000

b. Click on the “New Chat” option in the menu bar on the top left of the website.

c. Wait for the website to generate a new conversation.

- **Expected Results:**

a. The website will recognize that the “New Chat” option was pressed.

b. A new conversation will be appended below the previous conversation found in the “Recent Chats” in the menu bar.

c. There will be a transition from the current conversation with AMI to the new conversation created.

- **Post-Steps:**

a. Close the https://localhost:8000 tab.

b. Close AMI project.

- **Searching for a conversation:**

- **Instructions:**

a. Run the AMI project and navigate to https://localhost:8000

b. Create 5 new conversations following the “Creating a new conversation” test scenario and enter a different prompt in each using the “Creating a valid prompt” test scenario.

c. Click on the “Search Chats” option in the menu bar on the top left of the website.

d. Type the name of a conversation found in the “Recent Chats” in the menu bar.

e. Press the “Enter” on your keyboard or click on the submit button to the right of the text input field.

f. Click on the conversation.

g. Wait for the website to load the conversation.

- **Expected Results:**

a. The website will generate new conversations in accordance with the “Expected Results” section in the “Creating a new conversation” test scenario.

b. AMI will respond to the prompts in accordance with the “Expected Results” section in the “Creating a valid prompt” test scenario.

c. AMI will automatically generate a name to the 5 conversations according to the given prompts. These names will replace the placeholder text in the conversations found in the "Recent Chats" in the menu bar.

d. The website will recognize that the "Search Chats" option was pressed.

e. The text submitted by the user from the "Search Chats" option will be compared with the current names of the conversations found in the "Recent Chats" in the menu bar.

f. All other conversation names will disappear from the "Recent Chats" in the menu bar.

g. The closest conversation name that is relative to the submitted text will be shown in the "Recent Chats" in the menu bar.

h. There will be a transition from the current conversation with AMI to the chosen conversation.

- **Post-Steps:**

a. Close the https://localhost:8000 tab.

b. Close AMI project.

- **Deleting a conversation:**

- **Instructions:**

a. Run the AMI project and navigate to https://localhost:8000

b. Create a new conversation following the "Creating a new conversation" test scenario and enter a prompt using the "Creating a valid prompt" test scenario.

c. Click on the edit option shown as an edit icon to the right of the name of the conversation.

d. Click the "Delete" option.

e. Wait for the website to delete the conversation.

- **Expected Results:**

a. The website will generate a new conversation in accordance with the "Expected Results" section in the "Creating a new conversation" test scenario.

b. AMI will respond to the prompt in accordance with the "Expected Results" section in the "Creating a valid prompt" test scenario.

c. The website will recognize that the "Delete" option was pressed for the given conversation.

d. Conversation will be deleted, and the name associated with it will be deleted from the "Recent Chats" in the menu bar.

- **Post-Steps:**

a. Close the https://localhost:8000 tab.

b. Close AMI project.

3. **Settings:**

- **Changing the appearance to light mode:**

  - **Instructions:**

a. Run the AMI project and navigate to https://localhost:8000

b. Click on the "Settings" option on the bottom left of the website.

c. Click on the drop-down menu to the right of the "Appearance" option.

d. Select "light mode" option.

e. Wait for the website to convert to "light mode".

  - **Expected Results:**

a. The website by default registers the mode the user's system has.

b. The website will recognize that the "light mode" option was pressed.

c. The website will convert from "system mode" to "light mode".

  - **Post-Steps:**

a. Close the https://localhost:8000 tab.

b. Close AMI project.

- **Changing the appearance to dark mode:**

  - **Instructions:**

a. Run the AMI project and navigate to https://localhost:8000

b. Click on the "Settings" option on the bottom left of the website.

c. Click on the drop-down menu to the right of the "Appearance" option.

d. Select "dark mode" option.

e. Wait for the website to convert to “dark mode”.

- **Expected Results:**
    a. The website by default registers the mode the user’s system has.
    b. The website will recognize that the “dark mode” option was pressed.
    c. The website will convert from “system mode” to “dark mode”.
- **Post-Steps:**
    a. Close the https://localhost:8000 tab.
    b. Close AMI project.

UAT Outcome:

- Accepted [ ]
- Failed [ ]

Tester Signature:

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